

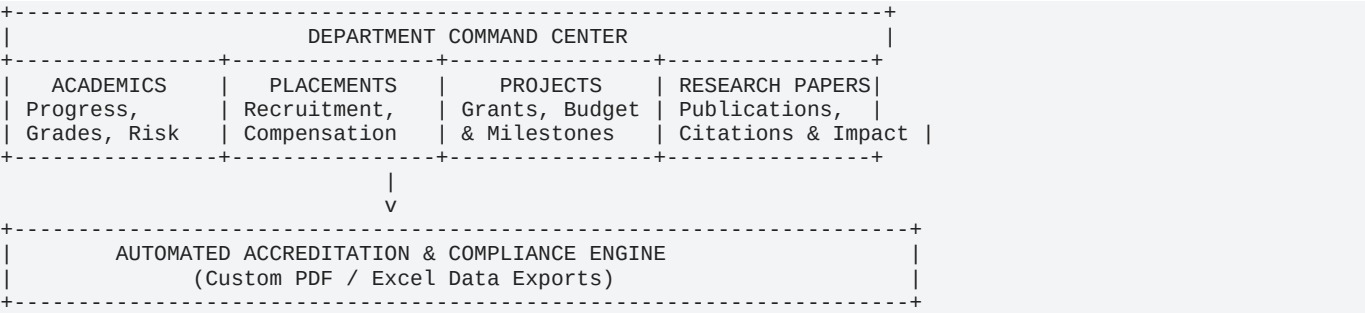
Product Requirements Document (PRD)

Department Analytics Platform

1. Product Overview & Purpose

The **Executive Department Analytics & Compliance Platform** is an enterprise management system designed for Academic Department Heads, Deans, and Executive Leadership. It centralizes disparate departmental operations—academics, recruitment, research grants, and scholarly publications—into a unified, real-time command center.

Currently, departmental metrics exist in isolated silos (LMS, placement drives, spreadsheets, external databases). This platform consolidates those feeds to enable **data-driven academic governance, proactive student intervention, and automated institutional accreditation reporting** (e.g., NAAC, NBA, NIRF, ABET).



2. Core User Personas & Workflows

Persona	Primary Goal	System Interaction & Usage
Department Head / Executive	High-level oversight, strategic decision-making, and audit compliance.	Consumes high-level KPIs, monitors risk alerts, triggers department-wide interventions, and exports compliance reports.
Faculty Member	Course delivery, student mentoring, and scholarly contributions.	Inputs grades, weekly syllabus completion, research paper submissions, and grant updates.
Placement Coordinator	Student career tracking and recruiter management.	Uploads offer letters, updates drive schedules, logs interview feedback, and tracks unplaced cohorts.
Department Admin / Clerk	Data entry and bulk uploads.	Uploads historical CSV/Excel data, manages batch rosters, and maintains master records.

3. Module Breakdown & Operational Specifications

Module 1: Academics & Student Performance

Purpose

To provide complete visibility into teaching quality, course progression, and student performance while automatically detecting academic risk before final examinations.

ACADEMIC PERFORMANCE OVERVIEW			Semester: Even 2026	
PASS RATE	AVG CGPA	AT-RISK	CGPA DISTRIBUTION	
88.4%	7.42	14	[==== 9.0+] (12%)	
(+3.2% YoY)	(+0.18)	Students	[===== 8.0-8.9] (34%)	
			[===== 7.0-7.9] (41%)	
			[=== <7.0] (13%)	
EARLY WARNING RADAR				
△ CS-301 (Data Structures): 8 Students attendance <60% & failed Quiz 1				
△ EC-202 (Signals): Syllabus completion lagging by 2 weeks				

Feature Specifications & Operational Usage

- **Department Pass & Grade Analytics:**
- **What We Build:** Interactive grade distribution curves (CGPA/GPA bell curves), batch-over-batch pass percentages, and subject-wise failure rates.
- **How It Is Used:** Department leadership reviews these charts during mid-semester reviews to evaluate grading consistency across subjects.
- **Early Warning Radar (Risk Detection):**
- **What We Build:** An automated rules engine that flags students falling below attendance thresholds (e.g., under 75%) or failing internal assessments.
- **How It Is Used:** Leadership identifies at-risk students 4 to 6 weeks before final exams and assigns them to faculty advisors for remedial tutoring.
- **Syllabus & Course Delivery Tracker:**
- **What We Build:** A weekly progress bar comparing planned vs. actual syllabus completion reported by course instructors.
- **How It Is Used:** Ensures all sections of a course maintain identical teaching velocity, preventing end-of-semester rushes.

Module 2: Career & Placements Tracking

Purpose

To monitor recruitment velocity, compensation dynamics, and student employability, enabling targeted interventions for unplaced students.

PLACEMENT & RECRUITMENT STATS				Batch: 2022-2026	
PLACED %	HIGHEST CTC	MEDIAN	OFFER	VELOCITY	
82.5%	\$42 PLA	\$12.5L	30	*	
(165/200)	(Product)	(+8%YOY)	20	*	*
			10	*	*
			0+		
UNPLACED COHORT DRILL-DOWN			Aug Sep Oct Nov		
• 35 Unplaced Students Core Skills: Java (18), Python (12), Core (5)					
• Recommended Action: Organize targeted System Design bootcamp.					

Feature Specifications & Operational Usage

- **Recruitment & Compensation KPI Suite:**

- **What We Build:** Real-time counters for placement percentage, total offers, median CTC, mean CTC, and highest package, categorized by industry sector (Product, Service, Core Engineering).
- **How It Is Used:** Provides immediate status reports during management reviews and showcases department outcomes to prospective industry partners.
- **Unplaced Cohort Profiler:**
- **What We Build:** A searchable, filterable dataset of unplaced students indexed by skill tags, academic backlogs, and preferred job roles.
- **How It Is Used:** Placement coordinators filter unplaced students skilled in specific technologies (e.g., Python, System Design) to invite them to matching company drives.
- **Industry Skill-Gap Heatmap:**
- **What We Build:** Aggregated feedback logs from visiting recruiters highlighting common student technical weaknesses during interviews.
- **How It Is Used:** Department leadership uses this data to adjust elective course offerings or schedule weekend skill bootcamps.

Module 3: Projects & Research Initiatives

Purpose

To track capstone, R&D, and sponsored project execution, milestones, and (where applicable) budget allocation across the department.

***Note:** Grant/budget fields are optional per-project, not mandatory — supports departments with limited or no active funded grants.*

RESEARCH GRANTS & INNOVATION		
ACTIVE PROJECTS	TOTAL GRANTS	GRANT BUDGET UTILIZATION
24	\$1.2M USD	Approved: [=====]
(14 Capstones,	(8 Industry,	Disbursed: [=====]
10 R&D)	2 Govt)	Utilized: [=====]
INTELLECTUAL PROPERTY & PATENTS		
• Filed this Year: 6 Patents Granted: 2 Commercialized: 1		
• Upcoming Deliverable: DST Grant Phase 2 Milestone due in 12 Days		

Feature Specifications & Operational Usage

- **Grant Financial Burn-Down Tracker:**
- **What We Build:** Financial visualization bars displaying approved budget vs. disbursed funds vs. actual spent amount per research project.
- **How It Is Used:** Project leaders and department heads ensure government/industry research funds are fully utilized before grant expiration dates.
- **Milestone & Deliverable Scheduler:**
- **What We Build:** An automated timeline flagging pending project deliverables, quarterly progress reports, and fund utilization certificates.
- **How It Is Used:** Prevents missed deadlines on sponsored R&D contracts, protecting departmental funding eligibility.
- **Patent & IP Pipeline Tracker:**
- **What We Build:** A funnel view of departmental intellectual property tracking items from provisional drafting to filing, grant, and commercial licensing.
- **How It Is Used:** Tracks annual innovation metrics required for institutional ranking frameworks (such as NIRF and QS rankings).

Module 4: Scholarly Publications & Citations

Purpose

To catalog faculty and student research output, evaluate journal quality, and measure overall departmental citation impact.

SCHOLARLY OUTPUT & CITATIONS		Calendar Year 2026	
PUBLICATIONS	TOTAL CITATIONS	JOURNAL QUALITY BREAKDOWN (Q-TIERS)	
48	1,240	[=====] Q1 (45%)	
(+12 vs 2025)	(h-index: 18)	[=====] Q2 (30%)	
		[=====] Q3/Q4 (25%)	
TOP FACULTY CONTRIBUTORS THIS QUARTER			
🏆 Dr. A. Sharma: 4 Q1 Papers (IEEE Transactions)			
🏆 Dr. R. Patel: 2 Patents, 3 Q2 Papers			

Feature Specifications & Operational Usage

- **Publication & Indexing Categorization Engine:**
- **What We Build:** A repository that auto-classifies papers into indexing categories (Scopus, Web of Science, IEEE, Springer) and journal quality tiers (Q1, Q2, Q3, Q4 based on SJR impact factors).
- **How It Is Used:** Leadership monitors whether the department is publishing in high-impact journals versus low-tier venues.
- **Citation & Impact Metrics:**
- **What We Build:** Aggregated displays of total departmental citations, overall h-index, and i10-index calculations.
- **How It Is Used:** Evaluates research growth year-over-year and provides verifiable data for university promotional materials and funding applications.
- **Faculty Contribution Ranking:**
- **What We Build:** An internal leaderboard showcasing top publishing faculty members per quarter or academic year.
- **How It Is Used:** Simplifies annual faculty performance evaluations (Appraisal/PBAS scores) and identifies candidates for research incentives.

4. Cross-Cutting Platform Capabilities

1. Unified Data Ingestion Engine (API + Manual/CSV)

- **What We Build:** A dual-entry architecture.
- **Automated Sync:** REST API connectors to pull data directly from institutional ERPs, LMS platforms, or external repositories (e.g., Scopus/ORCID APIs).
- **Manual & Bulk Upload UI:** A drag-and-drop CSV/Excel importer equipped with an instant validation engine that highlights missing or malformed rows before committing to the database.
- **How It Is Used:** Department staff can instantly import entire batch marksheets or placement lists via Excel, while automated connectors handle system-to-system syncing where available.

Open Question: Confirm feasibility of REST API connectors (ERP/LMS/Scopus) with IT before commit; if unavailable, v1 scope defaults to CSV/Excel ingestion only.

2. Role-Based Access Control (RBAC) & Security

- **What We Build:** Granular, permission-based access definitions.
- **Executive View:** Read-only access across all 4 modules and full report generation privileges.
- **Faculty View:** Access limited to updating assigned courses, logged research papers, and mentored projects.
- **Placement View:** Access restricted to managing recruitment drives, student offer entries, and company feedback logs.
- **How It Is Used:** Protects student privacy and prevents unauthorized edits while allowing each department member to update their respective domains.

3. Automated Compliance & Audit Report Generator

- **What We Build:** An asynchronous PDF and Excel generation engine that converts stored metrics into pre-formatted compliance tables matching accreditation criteria (e.g., NAAC, NBA, NIRF tables).
- **How It Is Used:** When an accreditation audit approaches, department leaders select the required reporting framework and generate completed documentation in seconds—eliminating weeks of manual spreadsheet consolidation.

4. Configurable Data Schema

- **What We Build:** Each module ships with a required core field set (mapped to accreditation reporting needs) plus an extensible custom-fields table, editable by Admins via the CMS without engineering involvement.
- **How It Is Used:** Allows departments to adapt record structures to their own data-management practices without a rebuild. Roles/RBAC remain as defined in Section 4.2.